Technology Architecture

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Purpose of this chapter

The technology architecture describes the platform on which the applications run and on which the data is stored. It covers the nodes, the system software, the technology services that the platform offers to the applications, and the artifacts in which data objects take physical form.

Like the application components, the system software is named with concrete open-source products. The technology services are the stable part of the reference. An NSI can realise the same service with a different product, or obtain it from a shared or hosted platform.

Position in the framework

In the TOGAF Architecture Development Method, the technology architecture is developed in Phase D. In ArchiMate, it is the technology layer, the lowest of the three core layers. It receives its requirements from the layers above and serves them.

PLACEHOLDER: Position of the Technology Architecture below the Data and Application Architecture

Direction	Relationship	Example in the SAF
From Motivation	Requirements are realised by technology elements	SAF-RQ07 is realised by the five storage zones; SAF-RQ02 by the Event Streaming Service
From Strategy	Technology services realise capabilities	The Object Storage Service realises IT infrastructure (3.2.5)
From Data Architecture	Artifacts realise data objects	The Statistics Zone realises *Statistics Dataset - SDMX (#4)*
From Application Architecture	Technology services serve application components	The Relational Database Service serves eleven application components

Direction	Relationship	Example in the SAF
Within the layer	Nodes host system software; system software realises technology services	Storage Nodes host Ceph RGW, which realises the Object Storage Service

How NSIs use this chapter

An NSI uses this chapter as a checklist of the technology services that the application landscape needs, and as one worked example of how to provide them with open-source software.

The reference platform is a Kubernetes cluster that can run on-premise or in a hybrid setup. An NSI that uses a public or government cloud can obtain several technology services as managed services, for example object storage, relational databases or secrets management. The technology service in the SAF stays the same; only the element that realises it changes. As in the application architecture, licences are listed, and SAF-CN01 applies.

VS-09 Infrastructure and Platform Enablement describes the business view of providing these platforms and services. This chapter describes what is provided.

Technology layers

The technology architecture contains 3 nodes, 15 system software elements, 12 technology services and 8 artifacts. They form four layers: nodes host system software, system software realises technology services, technology services serve the application components, and artifacts hold the data and the definitions.

PLACEHOLDER: ArchiMate view *Phase D - Technology Architecture: Layers*

Nodes

A **node** is a computational or physical resource that hosts other technology elements.

ID	Node	Description	Composed of
SAF-TN01	Kubernetes Platform Cluster	Container platform that hosts all application components. On-premise or hybrid, with separate namespaces for production, test and research (secure) workloads.	SAF-TN02, SAF-TN03; SAF-SW02, SAF-SW11, SAF-SW12, SAF-SW13, SAF-SW14, SAF-SW15
SAF-TN02	Compute Worker Nodes	General-purpose servers that run containers, Trino workers and statistical jobs.	SAF-SW01, SAF-SW03, SAF-SW05, SAF-SW06, SAF-SW07, SAF-SW08, SAF-SW09, SAF-SW10
SAF-TN03	Storage Nodes	Servers with local disks that form the object-storage cluster.	SAF-SW04

System Software

System software is software that provides an environment for storing, running and using other software or data.

ID	System software	Licence	Description	Hosted on	Realises technology service
SAF-SW01	Linux	GPL-2.0	Operating system for all nodes.	SAF-TN02	
SAF-SW02	Kubernetes	Apache-2.0	Container orchestration. Airflow KubernetesExecutor and Trino run on it.	SAF-TN01	SAF-TSV03
SAF-SW03	containerd	Apache-2.0	Container runtime.	SAF-TN02	SAF-TSV03
SAF-SW04	S3-Compatible Object Storage (Ceph RGW)	LGPL-2.1	Durable S3 API storage for all data zones. Alternative: MinIO (AGPL-3.0).	SAF-TN03	SAF-TSV01
SAF-SW05	Apache Iceberg Table Format	Apache-2.0	Open table format on S3: ACID, schema evolution, snapshots, branches and tags (used as steady-state quality versions).	SAF-TN02	SAF-TSV02
SAF-SW06	Apache Kafka	Apache-2.0	Event streaming backbone for registration events and lineage.	SAF-TN02	SAF-TSV04
SAF-SW07	PostgreSQL	PostgreSQL License	Relational store for Airflow, catalogs, Keycloak, .Stat Suite and the service desk.	SAF-TN02	SAF-TSV05
SAF-SW08	OpenSearch	Apache-2.0	Search and index for the metadata catalogue and log search.	SAF-TN02	SAF-TSV06

ID	System software	Licence	Description	Hosted on	Realises technology service
SAF-SW09	OpenBao	MPL-2.0	Secrets, certificates and transit encryption keys (used for pseudonymisation). Open-source fork of HashiCorp Vault.	SAF-TN02	SAF-TSV07
SAF-SW10	Observability Stack (Prometheus, OpenTelemetry, Grafana, Loki)	Apache-2.0 / AGPL-3.0	Metrics, traces, logs and alerting for the platform and the pipelines.	SAF-TN02	SAF-TSV08
SAF-SW11	Harbor	Apache-2.0	Container image registry with scanning and signing.	SAF-TN01	SAF-TSV09
SAF-SW12	Argo CD	Apache-2.0	GitOps delivery of platform and application releases.	SAF-TN01	SAF-TSV09
SAF-SW13	OpenTofu	MPL-2.0	Infrastructure as code for clusters, storage and networks.	SAF-TN01	SAF-TSV11
SAF-SW14	Velero	Apache-2.0	Backup and restore of cluster state and volumes.	SAF-TN01	SAF-TSV10
SAF-SW15	Traefik	MIT	Ingress controller with TLS termination.	SAF-TN01	SAF-TSV12

Technology Services

A **technology service** is the behaviour that the platform offers to application components and to other technology services.

ID	Technology service	Description	Realised by	Serves	Realises
SAF-TSV01	Object Storage Service	S3 API for the landing, raw, standardized, processed, statistics and released zones.	SAF-SW04	SAF-AC02, SAF-AC03, SAF-AC05, SAF-AC08, SAF-AC17; SAF-TSV02	3.2.5 IT infrastructure

ID	Technology service	Description	Realised by	Serves	Realises
SAF-TSV02	Table Storage Service	Iceberg tables with snapshots, branches and tags on object storage.	SAF-SW05	SAF-AC05, SAF-AC06, SAF-AC07, SAF-AC08	3.2.5 IT infrastructure
SAF-TSV03	Container Orchestration Service	Scheduling, scaling and isolation of containers; jobs for statistical workloads.	SAF-SW02, SAF-SW03	SAF-AC04, SAF-AC05, SAF-AC08, SAF-AC17	3.2.5 IT infrastructure
SAF-TSV04	Event Streaming Service	Durable events for *Registered* signals and lineage.	SAF-SW06	SAF-AC03, SAF-AC04, SAF-AC11	SAF-RQ02
SAF-TSV05	Relational Database Service	Managed PostgreSQL instances.	SAF-SW07	SAF-AC04, SAF-AC06, SAF-AC11, SAF-AC12, SAF-AC13, SAF-AC15, SAF-AC16, SAF-AC18, SAF-AC21, SAF-AC22, SAF-AC23	
SAF-TSV06	Search and Index Service	Search over metadata and logs.	SAF-SW08	SAF-AC11, SAF-AC16	
SAF-TSV07	Secrets and Key Management Service	Secrets, certificates and transit keys.	SAF-SW09	SAF-AC03, SAF-AC04, SAF-AC05, SAF-AC18	3.3.6 IT security services
SAF-TSV08	Observability Service	Metrics, logs, traces and alerts.	SAF-SW10	SAF-AC04, SAF-AC05, SAF-AC20	3.2.3 Technical management
SAF-TSV09	Image Registry and Delivery Service	Trusted images; together with Argo CD it delivers all releases (GitOps).	SAF-SW11, SAF-SW12	SAF-TSV03	
SAF-TSV10	Backup and Recovery Service	Backup of cluster state and persistent volumes.	SAF-SW14	SAF-TSV01, SAF-TSV05	
SAF-TSV11	Infrastructure Provisioning Service	Reproducible infrastructure from code.	SAF-SW13	SAF-TSV03	
SAF-TSV12	Ingress and TLS Service	Secure external access to portals and APIs.	SAF-SW15	SAF-AC13, SAF-AC14, SAF-AC15, SAF-AC16, SAF-AC17, SAF-AC20	

Artifacts

An **artifact** is a piece of data that is used or produced by software, such as a file, a table or a repository. In the SAF, artifacts are the physical form of the data objects. All artifacts are stored on the S3-compatible object storage (SAF-SW04).

The five zones are the physical steady states. Each zone is an Iceberg namespace, and each registered version in it is an immutable, tagged snapshot (SAF-RQ07).

ID	Artifact	Description	Realises
SAF-AF01	Raw Zone (S3 bucket + Iceberg namespace `raw`)	Landing prefix plus registered raw tables. Write-once; BIV classification tags; retention per agreement.	SAF-DO01, SAF-RQ07
SAF-AF02	Standardized Zone (namespace `std`)	Pseudonymised standardized tables that conform to the classifications.	SAF-DO02, SAF-RQ07
SAF-AF03	Processed Zone (namespace `proc`)	Edited and enriched microdata with quality-version tags and edit lineage.	SAF-DO03, SAF-RQ07
SAF-AF04	Statistics Zone (namespace `stat`)	SDMX-shaped tables (dimension, measure and attribute columns per DSD) with accuracy attributes. Approved products as dataflows.	SAF-DO04, SAF-RQ07
SAF-AF05	Released Zone (namespace `rel`)	Disclosure-controlled SDMX-shaped tables and the published SDMX messages.	SAF-DO06, SAF-RQ07
SAF-AF06	SDMX Messages (SDMX-ML, SDMX-JSON, SDMX-CSV)	Files and API responses in SDMX formats, generated from the statistics and released zones.	SAF-DO04, SAF-DO06
SAF-AF07	Airflow DAG Repository (Git)	DAG definitions, one set per stream, reviewed and released through GitLab.	SAF-DO18
SAF-AF08	dbt Project (Git)	Versioned SQL models, tests and documentation per stream.	SAF-DO18